Practice thoroughly to ensure that each individual component is delivered within exactly 2 minutes.

Student Must add a professional photo to this cage

AI-Based Multi-Modal Food Spoilage Prediction and Quality Assessment

Specialization: Information Technology (IT)

IT23348820 | Maddumage M. S. | J26-IT-333

AI-Based Multi-Modal Food Spoilage Prediction & Quality Assessment

Component 1 of 2: Research Problem, Gap, Proposed Solution & Objectives (J26-IT-333)

1. Research Problem & Impact

The Problem in Restaurants:

- Food spoilage in perishable inventory causes massive food waste and financial loss in restaurant SMEs.
- Manual sensory inspection is inconsistent, subjective, and cannot detect early micro-decay.
- Delayed detection leads to health code violations, customer illness, and carbon emissions.

Limitations of Existing Systems:

- Standalone monitoring: either vision-only (surface) or IoT gas-only (chamber) — no fusion.
- Generic classification: binary 'Fresh vs. Spoiled' without quantitative remaining shelf-life.
- Single model used across all produce without category-specific biochemical profiles.

2. Proposed Solution & Research Gap

Our Multi-Modal Solution:

- Combines continuous IoT chamber sensing (Temperature, Humidity, NH3, CO2, VOC) with Computer Vision.
- 42-dimensional CV feature vector: color moments, biological pigment degradation (NDVI, anthocyanin), texture & morphology.
- Weighted Multi-Modal ML Fusion: Random Forest / XGBoost regressor & classifier.

The Research Gap Addressed:

- No existing restaurant system fuses environmental IoT and optical CV with food-category profiles.
- We introduce category-specific modeling for Fruits, Vegetables, Meat, Fish, and Dairy products.
- Predicts quantitative Remaining Shelf-Life (hours) with 90% confidence interval, not just binary status.

3. Component Research Objectives

Main Research Objective:

Develop a predictive food spoilage detection system using IoT sensor data and machine learning to estimate remaining shelf-life in restaurant storage environments.

Component Tasks & Milestones:

1. IoT Sensor Module: ESP32 telemetry with DHT22, MQ-137 (NH3), MQ-135 (CO2), and SGP30 (VOC).
2. CV Quality Module: OpenCV surface analysis, pigment index extraction, ripeness stage & mould defect segmentation.
3. Category-Specific Models: Calibrated profiles across 47 produce species with 4-stage spoilage curves.
4. Multi-Modal Fusion Engine: Predict RSL hours, spoilage probability %, and 4-tier risk scores.
5. Dashboard Integration: Real-time REST API integration with alert threshold notifications.

AI-Based Multi-Modal Food Spoilage Prediction & Quality Assessment

Component 2 of 2: Technical Workflow, System Architecture & Prototype Contribution

System Workflow & Data Pipeline

1. Inputs (Heterogeneous Multi-Modal Streams):

- Vision: Food images (camera upload, ESP32-CAM stream) + food category label.
- IoT Telemetry: Temperature (°C), Humidity (% RH), NH3 (ppm), CO2 (ppm), VOC (ppm).

2. Processing Pipeline (<100ms inference on CPU):

- CV Engine: 42-dim vector (LAB/HSV color moments, NDVI, anthocyanin, Hu moments).
- Multi-Modal Fusion: 60% CV optical + 40% IoT microclimate weighting.

3. Intended Outputs:

- Quantitative Remaining Shelf-Life (RSL in hours) with 90% confidence interval.
- Spoilage Probability (%) & 4-Tier Risk Score: Low, Medium, High, Critical.
- Dynamic Storage Recommendations: adjust chiller temp/RH, prioritize for prep.

User Requirements & Technical Novelty

User Requirements Addressed:

- UR-1: Automatic produce species identification & ripeness verification.
- UR-2: Real-time risk alerts for kitchen staff prior to visible degradation.
- UR-3: Produce knowledge base with 4-stage spoilage progression for 47 species.

Key Technical Novelties:

1. Multi-modal sensor-vision fusion tailored specifically for restaurant storage microclimates.
2. Biological pigment index analysis (NDVI, chlorophyll degradation) beyond standard RGB.
3. Edge-deployable CPU-only ML pipeline (FastAPI + React + ESP32) without cloud GPU cost.

📸 Working Prototype Demonstration

[SCREENSHOT PLACEHOLDER: Insert Actual UI Screenshots Here]

- React 18 Vision Inspector with bounding boxes & pigment diagnostics
- Live ESP32 IoT Sensor Telemetry & Food Spoilage Risk Gauge
- Produce Encyclopedia modal with 47 calibrated fruits & vegetables

💬 Genuine User Feedback & Evaluation

[USER FEEDBACK PLACEHOLDER: Insert Pilot Evaluation Feedback Here]

- Feedback from restaurant kitchen managers, chefs, and inventory staff
- Qualitative review on alert usability, dashboard response time, and accuracy
- Documented user acceptance and operational recommendations for PP2

AI & IoT-Based Smart Restaurant Management System (J26-IT-333)

Commercialization, Market Feasibility & Environmental Sustainability (Overall Project)

Overall 4-Component Ecosystem: C1 Demand Prediction | C2 Kitchen Staff Optimization | C3 Food Spoilage (This Component) | C4 Smart Waste Bin

Customer Persona & Uniqueness

Customer Persona:

- Target: SME Restaurant Owners, Kitchen Operations Managers, Executive Chefs.
- Pain Points: 15–25% inventory loss from spoilage, inaccurate preparation quantities, unpredictable dining peaks.
- Goal: Minimize food waste, reduce ingredient costs, and maintain hygiene audit compliance.

What Makes Solution Unique:

- Unified Ecosystem: Demand forecasting, kitchen scheduling, spoilage prediction, and automated waste tracking on a single dashboard.
- Closed Feedback Loop: Spoilage & waste data feed directly back into purchasing models.

Market Size & Business Model

Market Opportunity:

- Global Food Waste Management Market: Estimated multi-billion dollar sector driven by sustainability mandates (SDG 12/13).
- **[PLACEHOLDER: Insert verified market size / TAM-SAM-SOM figures with formal citation]**

Commercialization Strategy:

- Tiered B2B SaaS: Monthly subscription per kitchen outlet (Basic, Pro, Enterprise).
- Modular IoT Hardware Kit: Low-cost plug-and-play ESP32 sensor pods sold at margin.
- API Licensing: White-label integration with existing POS & ERP restaurant systems.

Sustainability & Cost Recovery

Environmental & Social Impact:

- Environmental: Directly reduces organic waste in landfills, lowering methane/GHG emissions.
- Energy Efficient: Edge microcontrollers (ESP32) operate on ultra-low power.
- Social: Prevents foodborne illness by intercepting microbially degraded ingredients.

Cost Recovery for Restaurants:

- Direct Waste Savings: Target 15–20% reduction in food spoilage losses provides rapid ROI.
- Estimated payback period: 3–6 months based on average SME kitchen procurement savings.
- **[PLACEHOLDER: Insert formal financial payback projection table for PP2]**